

Ref	Component	Description	Qty
U1	ESP32 Module	ESP32-WROOM-32 WiFi/Bluetooth module	1
U2	USB-to-UART IC	CP2102N-A02-GQFN28R USB to Serial Converter	1
U3	Sensor	DHT22 Temperature & Humidity Sensor	1
U4	Voltage Regulator	3.3V LDO regulator (e.g., NCP1117-3.3 or equivalent)	1
J1	USB Connector	USB Type-C connector (power + data input)	1
LED1	Power LED	Power indicator LED (SMD)	1
LED2	User LED	User controllable/status LED	1
R1–R9	Resistors	Current limiting / pull-up / pull-down resistors (10 kΩ mix)	9
SW1	Push Button	BOOT button (GPIO0 boot mode selection)	1
SW2	Push Button	RESET (EN) button for ESP32 reset	1
Q1	Transistor	Auto-reset control transistor (USB-UART DTR/RTS control)	1
Q2	Transistor	Auto-reset control transistor (EN/BOOT handling)	1
J2–J	Pin Headers / Connectors	GPIO expansion connectors (SENSOR, IO, power rails, etc.)	2–3